MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 7AGP19

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: A 7-1/2 inch (190.5 mm) diameter, single beam, electrostatic focus and deflection cathode-ray tube having a special deflection structure for minimum deflection defocusing.

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	ed	Rg	Zd	Ehk
Unit:	V	Vdc	Vdc	Vdc	v	Meg	Meg	Vdc
Maximum:	6.9	0	3,500	10,000	2,000	1.5	5.0 (Note 1)	±180
Minimum:	5.7	-300	---	5,000	---	---	---	---
TEST CONDITION: (Note 2)	6.3	Adjust	Focus	8,000	---	---	---	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
(D)	Qualification inspection						
5101	Face tilt	-		---	---	---	---
5216	Cathode illumination	-		---	---	$10x10^{-6}$	ftL
5248	Deflection - factor uniformity	-		---	---	1	%
---	Deflection defocusing	5		---	---	1:1.5	Ratio
1331	Direct - interelectrode capacitance	-	Cathode to all	Ck	---	10.5	pF
			Grid 1 to all	Cg1	---	11.5	pF
			D1 to D2	C1D2	---	5.2	pF
			D3 to D4	C3D4	---	2.3	pF
			D1 to all	CD1	---	12.5	pF
			D2 to all	CD2	---	12.5	pF
			D3 to all	CD3	---	8.0	pF
			D4 to all	CD4	---	8.0	pF
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		---	---	---	---
	Quality conformance inspection, part 1	8					
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	3, 7	Beam current = 2 µAdc	---	---	---	---
5101	Angle between traces	-		---	89	91	Degrees
5106	Bulb, screen, and faceplate quality	4		---	---	---	---
5223	Modulation	3	Beam current = 2 µAdc	ΔEc1	---	22	Vdc
5226	Line width "A" (electrostatic deflection)	3	Beam current = 2 µAdc	Width	---	0.26	mm
5231	Spot position (electrostatic deflection)	-		---	---	15	mm
5231	Spot displacement (leakage)	-		---	---	10	mm
5241	Grid cutoff voltage	-		Eco	-105	-175	Vdc
5246	Focusing voltage at cutoff	-		Eb1	2,000	2,950	Vdc
5248	Deflection factor	-	+1D2	DF	175	195	Vdc/in.
5248	Deflection factor	-	+3D4	DF	170	186	Vdc/in.
5103	Pattern distortion	-		1D2	---	1/2	%
				3D4	---	1	%
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-		Ib1	-15	+10	µAdc
5101	Base alinement (electrostatic types)	-	+ID2, index pin	---	---	---	---
5216	Stray light emission	6	Eb2 = 10,000 Vdc	---	---	---	---
5226	Line width "B" (electrostatic types)	3	Beam current = 2 µAdc	Width	---	0.28	mm
5251	Heater - cathode leakage current	-		Ihk	---	15	µAdc
5221	Screen (P19)	-		cBf D0.1 D1.0 D1.0-D10	575 410 335 75	--- --- --- 120	cB cB cB cB
1101	Secureness of base, cap, or insert	-		---	---	---	---
(D) 1111	Base pin solder depth	-		---	---	---	---
(D) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test (1)	3	Group C; beam current = 2.0 µAdc; Eb2 = 10,000 Vdc	t	500	---	hrs
---	Life-test (1) end points:						
5223	Modulation	3	Beam current = 2.0 µAdc	ΔEc1	---	22	Vdc
5226	Line width "A"	3	Beam current = 2.0 µAdc	Width	---	0.26	mm
5226	Line width "B"	3	Beam current = 2.0 µAdc	Width	---	0.38	mm
---	Life test (2)	3	Group C; beam current = 2.0 µAdc; Eb2 = 10,000 Vdc	t	1,000	---	hrs
---	Life-test (2) end points:						
5223	Modulation	3	Beam current = 2.0 µAdc	ΔEc1	---	44	Vdc
5226	Line width "A"	3	Beam current = 2.0 µAdc	Width	---	0.52	mm
5226	Line width "B"	3	Beam current = 2.0 µAdc	Width	---	0.76	mm

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NOTES:

1. It is recommended that the deflection electrode circuit resistances be approximately equal.
2. The P19 screen can be permanently damaged if the current density is permitted to rise too high. To prevent burning, minimum beam current densities shall be employed.
3. Beam current is measured by applying a sufficient voltage to either D1 or D2 to deflect a 1 by 1 inch (25.4 by 25.4 mm) raster until the raster cannot be seen on the screen. Grid No. 1 bias voltage is adjusted so that either D1 or D2 will collect 2 microamperes.
4. No dead spot of diameter greater than 1 millimeter shall be permitted within the quality area, and in any 2-inch (50.8mm) circle within the quality area not more than one of a diameter greater than 0.7 millimeter, and not more than four of a diameter greater than 0.3 millimeter.
5. The ratio of the line width of a 6-inch (152.4 mm) long trace, adjusted for best center focus, to the line width measured on the faceplate axis when the trace is deflected perpendicularly $\pm$ 3 inches ($\pm$ 76.2 mm) from the face center by a balanced dc voltage, shall be not greater than the value specified herein.
6. With the tube enclosed in a lightproof container, with the potential specified herein applied, and with no deflecting fields applied, the tubes shall be biased to cutoff. Under these conditions, the entire face of the tube shall be examined for stray emission by an observer who has accommodated his eyes to viewing the face of the cathode-ray tube for at least 2 minutes through an aperture in the lightproof container. If stray emission occurs, scanning fields such as would be required to produce a 1 by 1 inch (25.4 by 25.4 mm) raster shall be applied, under which condition, the stray emission shall disappear completely.
7. Test shall be performed at the conclusion of the holding period.
8. Inspection shall be 100 percent.

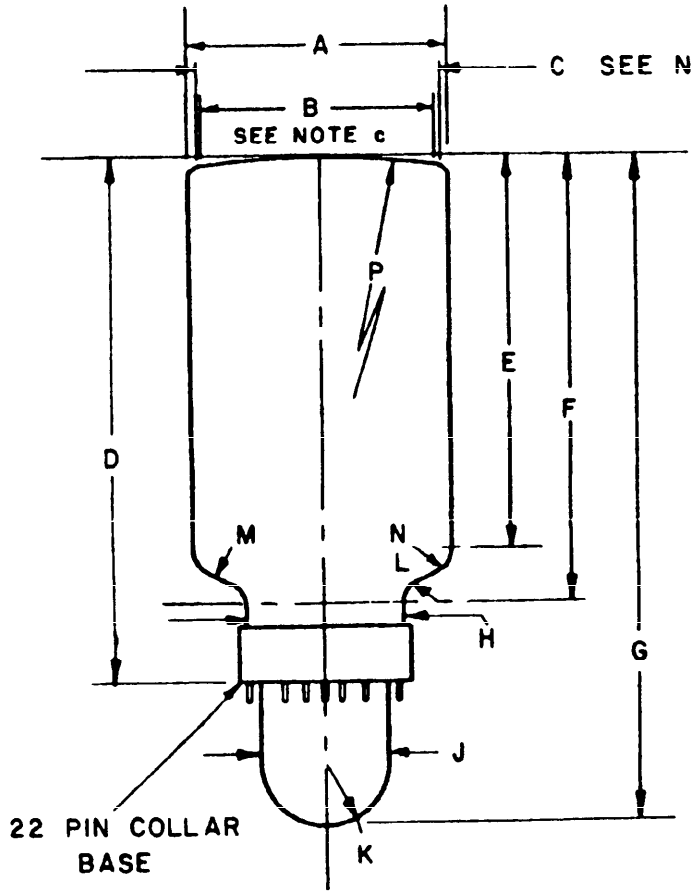
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Pin No.	Element
1	Accelerator
3	Deflector D1
5	Deflector D2
9	Focusing electrode
11	Cathode
12	Grid No. 1
14	Heater
15	Heater
19	Deflector D3
21	Deflector D4

NOTES:
a. Index pin to aline with 1D2 trace ±10°.
b. Minimum useful screen.
c. Minimum quality area.
d. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 7AGP19.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	7.312 (185.72)	7.688 (195.28)
B	6.750 (171.45)	
C	7.031 (178.59)	
D	14.500 (368.30)	15.000 (381.00)
E	10.500 (266.70)	11.000 (279.40)
F	12.250 (311.15)	12.750 (323.85)
G	18.688 (474.68)	19.000 (482.60)
H	4.312 (109.52)	4.500 (114.30)
J	3.469 (88.11)	3.656 (92.86)
Reference dimensions (see note d)		
K	.781 (19.84) R	
L	.500 (12.70) R	
M	5.063 (128.60) R	
N	.875 (22.22) R	
P	30.000 (762.00) R	

FIGURE 1. Outline drawing of electron tube type 7AGP19 - Continued.

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